

Day-Ahead Coordinated Scheduling of Water Distribution and Distributed Energy Systems: A Mixed-Integer Linear Programming Formulation

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Abstract—Water distribution pumping is a large, storage-backed electrical load whose flexibility is bounded by hydraulic service requirements on the water side and by feeder limits on the electricity side. This paper formulates the day-ahead coordinated scheduling of a water distribution network and a distributed energy system as a single mixed-integer linear program (MILP). Rather than proposing new component models, the formulation imports two published MILP building blocks. The water block is the MILPNet framework of Thomas and Sela, which represents nodal mass balance, piecewise-linear Hazen-Williams energy conservation, tanks, and the binary open/closed logic of pumps and valves. The energy block is the distributed-energy-system model of Morvaj *et al.*, restricted to its electrical part: the energy-hub electricity balance, storage, grid exchange, linearized AC power flow, and piecewise-linear branch-current limits. The heat balance and thermal technologies of that model are omitted because the water system interacts with the energy system only through electrical demand. The two blocks are joined by a single linear interface in which each pump's binary status draws a calibrated electrical power at its bus, so that pumping electricity is priced exactly once, through the grid procurement cost. Water and battery inventories are updated over every interval and must be replenished by the end of the horizon. A reproducible case study with a small water network and a two-bus feeder compares joint scheduling with an independently selected pumping schedule under identical tariffs and constraints. Joint scheduling lowers the daily procurement cost by moving pump operation out of the peak tariff block while leaving imported energy unchanged. Replaying the optimized schedules in EPANET and OpenDSS preserves pressure and voltage limits at report times but reveals a terminal water-reserve shortfall, which shows that nonlinear simulation checks are required beyond MILP feasibility.

Index Terms—Battery energy storage, day-ahead scheduling, linearized AC power flow, mixed-integer linear programming, pump scheduling, water distribution systems, water-energy nexus.

NOMENCLATURE

A. Sets and Indices

$\mathcal{T}, \bar{\mathcal{T}}$	Operating intervals $\{0, \dots, T-1\}$ and inventory boundaries $\{0, \dots, T\}$, index t .
$\mathcal{J}, \mathcal{K}, \mathcal{R}$	Water junctions, tanks, and reservoirs; $\mathcal{N} = \mathcal{J} \cup \mathcal{K} \cup \mathcal{R}$, index n (or i, j for link ends).
$\mathcal{L}^p, \mathcal{P}, \mathcal{V}$	Pipes, pumps, and valves; $\mathcal{L} = \mathcal{L}^p \cup \mathcal{P} \cup \mathcal{V}$, index l (p for pumps, v for valves).

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Code, network inputs, and the experiment scripts that regenerate the numerical section of this paper are available in the accompanying repository.

$\mathcal{B}, \mathcal{E}, \mathcal{G}$	Electrical buses, lines, and grid-connection buses $\mathcal{G} \subseteq \mathcal{B}$; indices b and l .
$\mathcal{P}_b$	Pumps electrically supplied from bus b .
A_{nl}^w, A_{bl}^e	Node-link incidence (+1 entering, -1 leaving, 0 otherwise) of the water and electrical graphs.

B. Parameters

Δt	Interval length (h); Δt_s the same in seconds.
$d_{n,t}^w$	Water demand at junction n (m^3/s).
$z_n, \underline{\pi}_n$	Junction elevation and minimum pressure head (m).
$H_{r,t}$	Reservoir total head (m).
K_l	Hazen-Williams resistance coefficient of pipe l .
$A_{k,t}, h_k^0$	Tank cross-sectional area (m^2) and initial head (m).
$h_k, \bar{h}_k$	Tank head bounds (m).
a_p, b_p, ν_p	Pump head-curve coefficients, $g_p(q) = a_p - b_p q^{\nu_p}$.
$\bar{q}_l, M_p$	Flow bound of link l (m^3/s) and pump disjunction constant (m).
$d_{b,t}^P, d_{b,t}^Q$	Non-pump active and reactive load at bus b (kW, kvar).
$\bar{p}_{b,t}^{\text{pv}}$	Available photovoltaic power (kW).
$\bar{e}_b, e_b^0$	Battery energy capacity and initial energy (kWh).
$\bar{p}_b^{\text{ch}}, \bar{p}_b^{\text{dis}}$	Battery charge and discharge power limits (kW).
$\eta_b^{\text{ch}}, \eta_b^{\text{dis}}$	Battery charge and discharge efficiencies.
$\bar{p}^g, \bar{p}^x$	Grid import and export capacities (kW).
R_l, X_l	Line series resistance and reactance (p.u.).
G_l, B_l	Line series conductance and susceptance (p.u.).
$\bar{i}_l$	Line current limit (p.u.).
$\underline{v}, \bar{v}$	Voltage magnitude bounds (p.u.); S_B power base (kVA).
$c_t^{\text{buy}}, c_t^{\text{sell}}$	Day-ahead purchase and sale prices (currency/kWh).
$\bar{P}_p, \eta_p$	Calibrated on-state electrical power (kW) and wire-to-water efficiency of pump p .

C. Decision Variables

$q_{l,t}, h_{n,t}$	Link flow (m^3/s) and nodal total head (m).
$\Delta h_{l,t}$	Pipe head loss (m).
$u_{p,t}$	Pump status, 1 when operating (binary).
$\gamma_{p,t}$	Pump head gain (m).
$y_{g,t}$	Gate-valve open status (binary).

$\alpha_{v,t}, \omega_{v,t}, \kappa_{v,t}$	PRV active, open, and closed status (binary).
λ, δ	Piecewise-linear weights (continuous) and segment selectors (binary).
$p_{b,t}^g, p_{b,t}^x$	Grid import and export (kW); $y_{b,t}^g, y_{b,t}^x$ their binary modes.
$r_{b,t}^g$	Reactive power drawn from the grid (kvar).
$p_{b,t}^{pv}$	Dispatched photovoltaic power (kW).
$p_{b,t}^{ch}, p_{b,t}^{dis}$	Battery charge and discharge at the AC bus (kW); $y_{b,t}^{ch}, y_{b,t}^{dis}$ their binary modes.
$e_{b,t}$	Battery energy at boundary t (kWh).
$p_{b,t}^{pump}$	Pump electrical demand at bus b (kW).
$f_{l,t}^P, f_{l,t}^Q$	Line active and reactive flow in the orientation of line l (kW, kvar).
$v_{b,t}, \theta_{b,t}$	Bus voltage magnitude (p.u.) and angle (rad).
$i_{l,t}^{re}, i_{l,t}^{im}$	Real and imaginary line-current components (p.u.).
$\phi_{l,t}, \chi_{l,t}$	Absolute values of $i_{l,t}^{re}$ and $i_{l,t}^{im}$; $\hat{\phi}_{l,t}, \hat{\chi}_{l,t}$ their piecewise-linear squares.

I. INTRODUCTION

WATER distribution systems are among the largest electricity consumers operated by municipalities, and pumping accounts for most of that consumption [1], [2]. Because elevated storage decouples the time at which water is pumped from the time at which it is consumed, pumping is a shiftable load: tanks can be filled before expensive tariff hours and drawn down afterwards. The same storage makes water utilities natural participants in day-ahead scheduling and demand response [3], [4]. The flexibility, however, is not free. On the water side, a pumping schedule must respect nodal mass balance, frictional head losses, pump characteristics, tank capacities, and minimum service pressures. On the electricity side, the pump load shares a feeder with other loads, photovoltaic (PV) generation, and batteries, so its preferred timing depends on voltage and thermal limits and on the timing of the other resources.

Two established mixed-integer linear programming (MILP) frameworks already describe each side of this problem. On the water side, Thomas and Sela [5] introduced MILPNet, a framework that represents mass and energy conservation, storage tanks, pumps, and valves within a MILP through piecewise linearization and big- M status logic, and demonstrated pump scheduling as one of its applications. On the energy side, Morvaj *et al.* [6] formulated the design and operation of a distributed energy system (DES) as an energy-hub MILP with integrated, linearized AC power-flow and branch-current constraints. Each framework treats the other side of the water–energy interface as exogenous: MILPNet prices pump operation with a fixed electricity cost, while the DES model takes electrical demand as a given profile.

This paper closes that gap by importing both models into one day-ahead scheduling problem. The water block is taken from MILPNet, and the energy block is taken from the DES model of [6] with its heat balance removed. Omitting heat

is not a simplification of convenience: the water network exchanges nothing with the thermal carrier, so the heat balance, the combined-heat-and-power, boiler, and heat-pump conversions, and the thermal storage of the DES model would only add variables that are unaffected by the coupling. The electrical part of the DES model is retained in full. The two blocks are joined by a single linear interface in which each pump’s binary status draws a calibrated electrical power at its assigned bus. The resulting problem remains a MILP, and pumping electricity is priced exactly once, through the grid procurement cost.

The contributions are as follows.

- 1) A day-ahead MILP for a single coordinating operator of a water distribution network and a distributed energy system, assembled from two published component models with an explicit mapping between each imported constraint and its source.
- 2) A coupling interface that keeps the joint problem linear, prices pump electricity exactly once, and accounts consistently for tank and battery inventories over every interval, including terminal replenishment of both.
- 3) A reproducible experimental protocol that isolates the value of joint scheduling from tariff and asset effects, together with an independent replay of the optimized schedules in nonlinear hydraulic (EPANET) and power-flow (OpenDSS) simulators.

The paper is organized as follows. Section II reviews related work. Section III states the scheduling problem. Sections IV and V present the imported water and energy blocks, and Section VI presents the coupling and the complete MILP. Section VII describes the implementation and experimental protocol, Section VIII reports the case study, Section IX discusses limitations, and Section X concludes.

II. RELATED WORK

Optimal operation of water distribution systems has a long history [7], [8]. Pump scheduling is the most common problem; solution approaches range from dynamic optimization [9] to MILP formulations with piecewise-linear hydraulics [10], Lagrangian decomposition [11], and convex programs for special network classes [12]. Menke *et al.* [3] showed that a MILP pump-scheduling model can quantify the demand response a water network can offer under time-varying prices. MILPNet [5] generalizes this line of work into a reusable framework: it reads an EPANET input file, formulates hydraulic elements, control rules, and status checks as MILP constraints, and compares its accuracy with EPANET on benchmark networks. The present paper adopts MILPNet as the water block and uses only its device models, since in a day-ahead problem the control rules are the decisions being optimized.

The water–power interface has been studied from the power-system side. Oikonomou *et al.* [4] schedule pumps as a demand-response resource against day-ahead prices, and Oikonomou and Parvania [13] integrate the energy flexibility of a water distribution system into day-ahead power-system scheduling. Zamzam *et al.* [14] formulate the joint optimal water–power flow problem and solve it in a distributed manner,

while Fooladivanda and Taylor [15] derive convex relaxations of the hydraulic constraints for energy-optimal pumping. Stuhlmacher and Mathieu [16] schedule pumps under water and power demand uncertainty subject to distribution-network constraints. Santhosh *et al.* [17] address the supply side of the nexus with co-production plants. These works propose their own hydraulic and electrical models. The present paper instead assembles the joint problem from two frameworks that were each validated on their own domain, which makes the origin of every constraint traceable and simplifies extension to additional water devices or energy technologies.

On the energy side, the energy-hub concept [18], [19] describes conversion and storage among carriers by constant efficiencies, and Evins *et al.* [20] extended it with operational constraints. Morvaj *et al.* [6] embedded the linearized AC power-flow model of Koster and Lemkens [21] in the hub MILP and added a piecewise-linear branch-current magnitude limit, so that grid constraints influence the optimized dispatch instead of being checked afterwards. Alternative linear or convex distribution-network models include the linear-programming approximation of AC power flow [22] and the DistFlow branch-flow model and its relaxations [23], [24]. The present paper keeps the formulation of [6] because it is a MILP with the same piecewise-linear machinery as MILPNet, so both blocks can be solved by one solver without conic constraints.

III. PROBLEM STATEMENT

A single operator schedules, one day ahead, a water distribution network and a distributed energy system that share an electrical feeder (Fig. 1). The horizon consists of T operating intervals $\mathcal{T}$ of length Δt and $T + 1$ inventory boundaries $\bar{\mathcal{T}}$. Water demands, non-pump electrical loads, PV availability, reservoir heads, and day-ahead prices are known for the horizon. The operator chooses pump statuses and flows, PV curtailment, battery charging and discharging, and grid import and export, subject to hydraulic and electrical network constraints, and pays for imported electricity at the day-ahead price. The objective is the daily electricity procurement cost net of export revenue; water and electricity service are constraints, not costs.

The decision structure is that of a centralized benchmark: all network data and forecasts are available to one decision maker, and no bidding, contract, or decentralized coordination mechanism is modeled. The intended use is twofold: as an operational day-ahead schedule for a utility that owns both systems, and as the reference against which decentralized coordination schemes can be measured.

Table I summarizes the constraints of the joint problem and their origin. Equations are numbered as they appear below; the source column refers to the equation numbers of [5] (T&S) and [6] (M).

IV. WATER DISTRIBUTION BLOCK

The water block is the MILPNet formulation of Thomas and Sela [5]. A directed graph contains junctions $\mathcal{J}$, tanks $\mathcal{K}$, reservoirs $\mathcal{R}$, and links $\mathcal{L}$ consisting of pipes $\mathcal{L}^P$, pumps $\mathcal{P}$,

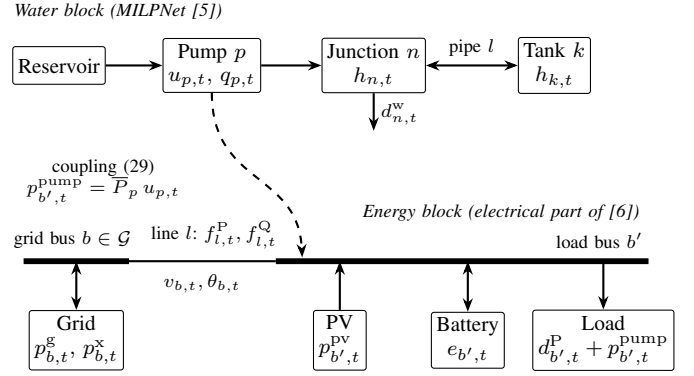


Fig. 1. Coupled day-ahead scheduling problem. The water block and the energy block are imported from published MILP models; the only shared constraint is the pump electrical demand at its supply bus.

TABLE I
CONSTRAINTS OF THE JOINT PROBLEM AND THEIR ORIGIN

Constraint	Source	Treatment in this paper
<i>Water block</i>		
Junction mass balance (1)	T&S (1)	As published
Pipe energy conservation (2)	T&S (2)	As published
Hazen-Williams loss (3)	T&S (3)	PWL over a per-pipe signed domain
Tank dynamics (7)	T&S (4)	Plus bounds and terminal reserve
Pump curve and status (10)–(12)	T&S (5)–(7)	Status 1 = on; closure slack eliminated
Gate valve (13)	T&S (8), (11)	As published
PRV states (14)	T&S (9), (12)	As published; not exercised
Tank-link checks, control rules	T&S (13)–(15)	Omitted: statuses are decisions
Pressure floor (4)	—	Added service constraint
<i>Energy block</i>		
Electricity balance (15)	M (1), (14)	Nodal form; pump load added
Heat balance	M (2)	Omitted: no thermal coupling
Storage continuity (17)	M (3)	Efficiencies applied once
Capacity and exclusivity (20)–(22)	M, Sec. 2.2	As described; terminal closure added
Linearized AC flow (23)–(24)	M (10), (11)	$U_0 = 1$ p.u.
Reactive balance (16)	M (15)	Grid supplies reactive power
Branch current (25)–(28)	M (25)–(31)	SOS2 interpolation; no objective weight
Cost (31)	M (4)	Electricity streams only
<i>Coupling</i>		
Pump demand (29), calibration (30)	—	This paper

and valves $\mathcal{V}$. Link orientation fixes the sign of flow; pipes and gate valves may carry flow in either direction, whereas pumps do not allow backflow. Reservoir heads follow their prescribed time series, $h_{r,t} = H_{r,t}$ for $r \in \mathcal{R}$.

A. Conservation Laws

Mass balance at every junction and interval requires that the net inflow equal the demand [5, Eq. (1)]:

$$\sum_{l \in \mathcal{L}} A_{nl}^w q_{l,t} = d_{n,t}^w, \quad n \in \mathcal{J}, t \in \mathcal{T}. \quad (1)$$

Demand is the sum over all demand categories evaluated with the network's time patterns; consumption is demand driven, so pressure-dependent demand and leakage are not represented.

Energy conservation along a pipe $l = (i, j)$ relates the head difference to the head loss [5, Eq. (2)],

$$h_{i,t} - h_{j,t} - \Delta h_{l,t} = 0, \quad l \in \mathcal{L}^p, \quad t \in \mathcal{T}, \quad (2)$$

and the head loss follows the Hazen–Williams relation [5, Eq. (3)],

$$\Delta h_{l,t} = \text{sgn}(q_{l,t}) K_l |q_{l,t}|^{1.852}, \quad K_l = \frac{10.67 L_l}{C_l^{1.852} D_l^{4.8704}}, \quad (3)$$

with pipe length L_l (m), diameter D_l (m), and roughness C_l . Equation (3) is nonlinear and nonconvex because of the bidirectional flow, and MILPNet replaces it by a piecewise-linear (PWL) interpolation of the signed curve over a bounded flow domain. A service constraint that MILPNet leaves to the user is added here as a minimum pressure head at every junction:

$$h_{n,t} - z_n \geq \underline{\pi}_n, \quad n \in \mathcal{J}, \quad t \in \mathcal{T}. \quad (4)$$

B. Piecewise-Linear Representation

All univariate nonlinearities in both blocks use the same construction. Given breakpoints (x_s, y_s) , $s = 0, \dots, S$, with strictly increasing x_s , the relation $y = \text{PWL}(x)$ is represented by nonnegative weights λ_s and segment binaries δ_s , $s = 0, \dots, S-1$ [25]–[27]:

$$x = \sum_{s=0}^S \lambda_s x_s, \quad y = \sum_{s=0}^S \lambda_s y_s, \quad \sum_{s=0}^S \lambda_s = \sum_{s=0}^{S-1} \delta_s = \zeta, \quad (5)$$

$$\lambda_s \leq \delta_{s-1} + \delta_s, \quad \delta_s \in \{0, 1\}, \quad (6)$$

where undefined end binaries are zero and ζ is an activation term. For pipes and lines $\zeta = 1$; for a pump $\zeta = u_{p,t}$, so that both the flow and the head gain vanish when the pump is off. Adjacent breakpoints only can carry positive weight, which is the special-ordered-set (SOS2) condition written with explicit binaries; the formulation is therefore independent of solver-specific SOS or indicator support. For the Hazen–Williams curve, the breakpoints span a signed domain $[-\bar{q}_l, \bar{q}_l]$ with zero as a breakpoint; following MILPNet, the domain is set from an extended-period simulation of the network so that the segments cover the actual operating range. The interpolation is an equality surrogate of the nonlinear relation, not a relaxation with a guaranteed hydraulic error bound; the case study reports the resulting discrepancies through replay.

C. Tanks and Inventory

For a cylindrical tank k the state is its total head, which evolves with the net inflow over each interval [5, Eq. (4)]:

$$h_{k,t+1} = h_{k,t} + \frac{\Delta t_s}{A_k} \sum_{l \in \mathcal{L}} A_{kl}^w q_{l,t}, \quad k \in \mathcal{K}, \quad t \in \mathcal{T}, \quad (7)$$

$$h_{k,0} = h_k^0, \quad \underline{h}_k \leq h_{k,t} \leq \bar{h}_k, \quad k \in \mathcal{K}, \quad t \in \bar{\mathcal{T}}, \quad (8)$$

$$h_{k,T} \geq h_k^0, \quad k \in \mathcal{K}. \quad (9)$$

Two points distinguish (7)–(9) from a naive discretization. First, the inventory is defined on all $T+1$ boundaries, so the withdrawal or filling decided in the last interval changes $h_{k,T}$

and cannot disappear from the accounting. Second, the terminal condition (9) preserves the initial reserve, which prevents apparent savings obtained by draining the tank; all compared schedules must use the same terminal requirement. MILPNet's tank-link status checks [5, Eq. (13)], which reproduce the automatic closing of a tank inlet at full or empty conditions in a simulator, are replaced by the hard bounds (8): a schedule that respects the bounds at every boundary does not trigger those checks in the discretized model, and whether it does so within an interval is a question for the replay of Section VIII.

D. Pumps

A fixed-speed pump $p = (i, j)$ adds head according to a head–flow curve when it operates [5, Eq. (5)]. MILPNet uses the quadratic form $a_p - b_p q^2$; EPANET's power-law form $g_p(q) = a_p - b_p q^{\nu_p}$ is used here so that single-point and three-point curves are covered by the same expression, and g_p is interpolated over $[0, (a_p/b_p)^{1/\nu_p}]$ with the construction of Section IV-B:

$$\gamma_{p,t} = \text{PWL}_p(q_{p,t}) \text{ with } \zeta = u_{p,t}, \quad p \in \mathcal{P}, \quad t \in \mathcal{T}. \quad (10)$$

MILPNet models the two states of a pump with a closure slack on the energy balance and a big- M pair [5, Eqs. (6)–(7)]. Eliminating the slack gives the equivalent form

$$0 \leq q_{p,t} \leq \bar{q}_p u_{p,t}, \quad (11)$$

$$-M_p(1 - u_{p,t}) \leq h_{j,t} - h_{i,t} - \gamma_{p,t} \leq M_p(1 - u_{p,t}), \quad (12)$$

for $p \in \mathcal{P}$, $t \in \mathcal{T}$. When $u_{p,t} = 1$ the downstream head equals the upstream head plus the curve gain; when $u_{p,t} = 0$ the flow and the gain are zero and the two end heads are decoupled. The status convention is the complement of MILPNet's, where $y_p^t = 0$ denotes an open pump; the convention used here makes the coupling term of Section VI a product of the status with a positive power. Finite flow and head bounds are needed both for physical interpretation and for valid constants M_p , which MILPNet sets just above the largest bound of the associated continuous variable.

E. Valves

Gate valves and pressure-reducing valves (PRVs) are retained from MILPNet for networks that contain them; the case study network has none, so these constraints are stated compactly. A gate valve $g = (i, j)$ with open status $y_{g,t}$ behaves like a pipe when open and disconnects its ends when closed [5, Eqs. (8), (11)]:

$$-\bar{q}_g y_{g,t} \leq q_{g,t} \leq \bar{q}_g y_{g,t}, \quad (13)$$

$$-M_g(1 - y_{g,t}) \leq h_{i,t} - h_{j,t} - \Delta h_{g,t} \leq M_g(1 - y_{g,t}),$$

with $\Delta h_{g,t}$ the PWL Hazen–Williams loss of the valve. A PRV $v = (i, j)$ with setting H_v^{set} is in exactly one of three states, active ($\alpha_{v,t}$), open ($\omega_{v,t}$), or closed ($\kappa_{v,t}$) [5, Eqs. (9), (12)]:

$$\begin{aligned} \alpha_{v,t} + \omega_{v,t} + \kappa_{v,t} &= 1, \quad 0 \leq q_{v,t} \leq \bar{q}_v(1 - \kappa_{v,t}), \\ |h_{j,t} - H_v^{\text{set}}| &\leq M_v(1 - \alpha_{v,t}), \quad h_{i,t} \geq H_v^{\text{set}} - M_v(1 - \alpha_{v,t}), \\ |h_{i,t} - h_{j,t}| &\leq M_v(1 - \omega_{v,t}), \quad h_{i,t} \leq H_v^{\text{set}} + M_v(1 - \omega_{v,t}), \\ h_{i,t} - h_{j,t} + \epsilon \kappa_{v,t} &\leq M_v(1 - \kappa_{v,t}), \end{aligned} \quad (14)$$

where each absolute value denotes the corresponding pair of linear inequalities and $\epsilon > 0$ replaces the strict inequality of the closed state, as in MILPNet. An active PRV holds its downstream head at the setting, an open PRV passes flow without additional loss, and a closed PRV blocks reverse flow.

F. Control Rules

MILPNet also formulates event-based and time-based control rules [5, Eqs. (14)–(15)], which make the MILP reproduce the rule-based behavior of an extended-period simulation. In the day-ahead scheduling problem the pump statuses are the decisions, so those rules are omitted; the rule-based schedule of the input file is used only for calibration (Section VI) and as an independent reference in the replay. MILPNet's pump-scheduling objective [5, Eq. (16)] contains a fixed energy cost per operating hour and a switching penalty. The energy cost is replaced by the electrical coupling of Section VI; the switching penalty is not included in the present implementation, but it is linear in $u_{p,t}$ and can be added without changing the problem class.

V. DISTRIBUTED ENERGY SYSTEM BLOCK

The energy block is the DES model of Morvaj *et al.* [6], whose energy-hub layer assigns conversion and storage technologies to meet electricity and heat demand while a linearized AC power-flow layer represents the distribution grid. Only the electrical part is imported. Each electrical bus $b \in \mathcal{B}$ hosts a hub with, at most, a PV generator, a battery, non-pump loads, and, if $b \in \mathcal{G}$, a grid connection. Conversion technologies with heat output and the thermal storage are absent, so the conversion-efficiency matrix of the hub reduces to the identity for PV and the heat balance [6, Eq. (2)] is dropped. The asset capacities are given, not designed.

A. Electricity Balance

The hub electricity balance [6, Eq. (1)] and the nodal active-power balance of the grid layer [6, Eq. (14)] are combined into one balance per bus, with the pump demand of Section VI added to the load:

$$p_{b,t}^g - p_{b,t}^x + p_{b,t}^{\text{pv}} + p_{b,t}^{\text{dis}} - p_{b,t}^{\text{ch}} + \sum_{l \in \mathcal{E}} A_{bl}^e f_{l,t}^P = d_{b,t}^P + p_{b,t}^{\text{pump}}, \quad b \in \mathcal{B}, t \in \mathcal{T}. \quad (15)$$

The reactive balance [6, Eq. (15)] is

$$r_{b,t}^g + \sum_{l \in \mathcal{E}} A_{bl}^e f_{l,t}^Q = d_{b,t}^Q, \quad b \in \mathcal{B}, t \in \mathcal{T}, \quad (16)$$

where PV, battery, and pump exchanges are at unity power factor and reactive load is supplied from the grid connection; $r_{b,t}^g = 0$ for $b \notin \mathcal{G}$. A non-unity pump power factor would add a term to (16) proportional to $p_{b,t}^{\text{pump}}$.

B. Storage, Generation, and Grid Exchange

Storage continuity [6, Eq. (3)] is written with charging and discharging measured at the AC bus, so that the efficiencies appear once, in the inventory equation, and not again in the bus balance:

$$e_{b,t+1} = e_{b,t} + \Delta t \left(\eta_b^{\text{ch}} p_{b,t}^{\text{ch}} - \frac{p_{b,t}^{\text{dis}}}{\eta_b^{\text{dis}}} \right), \quad t \in \mathcal{T}, \quad (17)$$

$$e_{b,0} = e_b^0, \quad 0 \leq e_{b,t} \leq \bar{e}_b, \quad t \in \bar{\mathcal{T}}, \quad (18)$$

$$e_{b,T} \geq e_b^0. \quad (19)$$

As for tanks, the inventory is tracked on all $T+1$ boundaries and the terminal energy must be replenished. PV can be curtailed within its availability,

$$0 \leq p_{b,t}^{\text{pv}} \leq \bar{p}_{b,t}^{\text{pv}}, \quad (20)$$

and, as required in [6], a storage cannot charge and discharge, nor can a bus import and export, in the same interval:

$$0 \leq p_{b,t}^{\text{ch}} \leq \bar{p}_{b,t}^{\text{ch}} y_{b,t}^{\text{ch}}, \quad 0 \leq p_{b,t}^{\text{dis}} \leq \bar{p}_{b,t}^{\text{dis}} y_{b,t}^{\text{dis}}, \quad y_{b,t}^{\text{ch}} + y_{b,t}^{\text{dis}} \leq 1, \quad (21)$$

$$0 \leq p_{b,t}^g \leq \bar{p}_{b,t}^g y_{b,t}^g, \quad 0 \leq p_{b,t}^x \leq \bar{p}_{b,t}^x y_{b,t}^x, \quad y_{b,t}^g + y_{b,t}^x \leq 1, \quad (22)$$

for $b \in \mathcal{B}$, $t \in \mathcal{T}$, with $p_{b,t}^g = p_{b,t}^x = 0$ for $b \notin \mathcal{G}$ and zero storage and PV bounds at buses without those assets. Battery degradation is not modeled.

C. Linearized AC Power Flow

Each line $l = (i, j)$ has series impedance $R_l + jX_l$ and series admittance $G_l + jB_l$, with $G_l = R_l/(R_l^2 + X_l^2)$ and $B_l = -X_l/(R_l^2 + X_l^2)$. Morvaj *et al.* adopt the linearization of [21], a first-order expansion of the AC power-flow equations about a flat voltage profile U_0 with small angle differences [6, Eqs. (10)–(11)]. With $U_0 = 1$ p.u. and the power base S_B ,

$$f_{l,t}^P = S_B [G_l(v_{i,t} - v_{j,t}) - B_l(\theta_{i,t} - \theta_{j,t})], \quad (23)$$

$$f_{l,t}^Q = S_B [-B_l(v_{i,t} - v_{j,t}) - G_l(\theta_{i,t} - \theta_{j,t})], \quad (24)$$

for $l \in \mathcal{E}$, $t \in \mathcal{T}$. Voltage magnitudes obey $\underline{v} \leq v_{b,t} \leq \bar{v}$, and the reference bus has $v = 1$ and $\theta = 0$. The model is balanced, lossless, and topology fixed: each line has one signed flow whose receiving-end value is its negative, and shunt charging, transformer tap changes, phase imbalance, and higher-order terms are absent. A solution of (23)–(24) is therefore not a certificate of AC feasibility, which is why the optimized injections are replayed in a nonlinear power-flow solver in Section VIII.

D. Branch-Current Limits

Distribution feeders are constrained by conductor ampacity as well as by voltage. Morvaj *et al.* linearize the branch current by the same expansion, obtaining real and imaginary parts that are linear in the voltage and angle differences [6, Eq. (25)]. At the expansion point these components coincide with the per-unit line flows of (23)–(24):

$$i_{l,t}^{\text{re}} = G_l(v_{i,t} - v_{j,t}) - B_l(\theta_{i,t} - \theta_{j,t}) = f_{l,t}^P/S_B, \quad (25)$$

$$i_{l,t}^{\text{im}} = B_l(v_{i,t} - v_{j,t}) + G_l(\theta_{i,t} - \theta_{j,t}) = -f_{l,t}^Q/S_B. \quad (26)$$

The magnitude requires the sum of the squared components. Following [6, Eqs. (26)–(31)], nonnegative auxiliary variables bound the absolute values,

$$\phi_{l,t} \geq \pm i_{l,t}^{\text{re}}, \quad \chi_{l,t} \geq \pm i_{l,t}^{\text{im}}, \quad (27)$$

their squares are interpolated on $[0, \bar{i}_l]$ with the construction of Section IV-B, and the interpolated squares are limited:

$$\hat{\phi}_{l,t} = \text{PWL}(\phi_{l,t}^2), \quad \hat{\chi}_{l,t} = \text{PWL}(\chi_{l,t}^2), \quad \hat{\phi}_{l,t} + \hat{\chi}_{l,t} \leq \bar{i}_l^2. \quad (28)$$

Two details differ from [6]. There, the absolute values are pushed toward the current components by a small objective weight, and the convexity of x^2 makes the SOS2 adjacency condition unnecessary. Here the adjacency binaries of (6) are kept, so the same interpolation code serves both blocks, and no objective perturbation is needed: increasing $\phi_{l,t}$ or $\chi_{l,t}$ beyond the absolute value can only tighten (28), so any feasible schedule remains feasible with $\phi_{l,t} = |i_{l,t}^{\text{re}}|$ and $\chi_{l,t} = |i_{l,t}^{\text{im}}|$. Because each secant lies above x^2 , the constraint is conservative for the magnitude of the linearized current, with a maximum interpolation excess of $\Delta x^2/4$ on a segment of width Δx ; finer segments reduce this conservatism at the cost of more binaries but do not correct the error of the underlying linearization.

VI. COUPLING AND THE DAY-AHEAD MILP

A. Pump Electrical Demand

The two blocks share one constraint. Each running pump draws a constant electrical power $\bar{P}_p$ at the bus that supplies it, and the demand at bus b is

$$p_{b,t}^{\text{pump}} = \sum_{p \in \mathcal{P}_b} \bar{P}_p u_{p,t}, \quad b \in \mathcal{B}, \quad t \in \mathcal{T}, \quad (29)$$

which enters the bus balance (15). The coefficient is calibrated from an extended-period EPANET simulation of the input network under its original controls, the same simulation that MILPNet uses to bound the PWL domains. With $q_{p,\tau}^{\text{sim}}$ and $\Delta h_{p,\tau}^{\text{sim}}$ the simulated flow and head gain of pump p , and $\mathcal{O}_p$ the set of simulation steps in which it runs,

$$\bar{P}_p = \frac{\rho g}{\eta_p} \cdot \frac{1}{|\mathcal{O}_p|} \sum_{\tau \in \mathcal{O}_p} q_{p,\tau}^{\text{sim}} \Delta h_{p,\tau}^{\text{sim}}, \quad (30)$$

with $\rho g = 9.81 \text{ kW}\cdot\text{s}/\text{m}^4$ for water. Equation (29) is linear in the existing pump status, so the coupled problem introduces no new nonlinearity or integrality. It is also the principal coupling approximation: the modeled electrical draw depends on whether the pump runs, not on the optimized flow and head. Its consequences are examined in Sections VIII and IX.

B. Objective

The DES objective of [6, Eq. (4)] is a weighted sum of input energy streams. With natural gas absent and the weights equal to day-ahead prices, it reduces to the electricity procurement cost net of export revenue:

$$\min C = \Delta t \sum_{t \in \mathcal{T}} \sum_{b \in \mathcal{G}} (c_t^{\text{buy}} p_{b,t}^g - c_t^{\text{sell}} p_{b,t}^x). \quad (31)$$

Pumping enters (31) only through (29) and (15), so it is charged once; the fixed per-hour pump cost of MILPNet's scheduling objective is removed to avoid double counting. No artificial water-flow cost is added.

C. Complete Problem and Properties

The day-ahead coordinated scheduling problem is

$$\begin{aligned} \min \quad & C \text{ in (31)} \\ \text{s.t.} \quad & \text{water block: (1)–(4), (7)–(14),} \\ & \text{energy block: (15)–(28),} \\ & \text{coupling: (29),} \\ & \text{PWL relations: (5)–(6), } u, y, \alpha, \omega, \kappa, \delta \in \{0, 1\}. \end{aligned} \quad (32)$$

Problem (32) is a MILP. Its binaries are the pump statuses, valve states, storage and grid modes, and the PWL segment selectors; all other variables are continuous. Three properties follow directly from the construction. First, tanks and batteries have the same accounting role: each can move consumption across time but neither can supply net energy or water through an unrecorded terminal withdrawal, because of (9) and (19). Second, with exact solves, fixing the pump statuses to any feasible schedule cannot decrease the optimal cost, so the difference between a fixed-schedule solve and the joint solve is a nonnegative measure of the value of coordination under identical assumptions; a violation of this ordering indicates unequal assumptions, a solver gap, or a reporting error rather than negative flexibility. Third, if the demand balance (1) is relaxed with slacks $s_{n,t}^{\pm}$ and a large penalty for diagnosis, any nonzero slack is a service violation and must be reported as such; the results below use the exact balance.

VII. IMPLEMENTATION AND EXPERIMENTAL PROTOCOL

A. Workflow

The formulation is implemented in Python with Pyomo [28], [29] as a shared model assembled from optional water and energy components (Fig. 2). EPANET input files are read through WNTR [30], and buses, loads, line parameters, and base quantities are extracted from an OpenDSS circuit [31]. A configuration file specifies the horizon, assets, tariffs, pump-to-bus assignments, PWL segment counts, bounds, and solver options, which separates the physical network description from the economic experiment. The workflow validates the configuration, runs the EPANET calibration simulation that supplies the PWL flow domains and the pump coefficients (30), builds the MILP, solves it, checks solver status and the presence of an incumbent before extracting a solution, and writes schedules, inventories, and summary metrics. The case study was solved with Gurobi 12.0.3 [32] through Pyomo 6.9.4 at a relative optimality gap tolerance of 10^{-4} ; the open-source solver HiGHS [33] is supported as well. Dependency versions and hashes of the source and input files are recorded with every run.

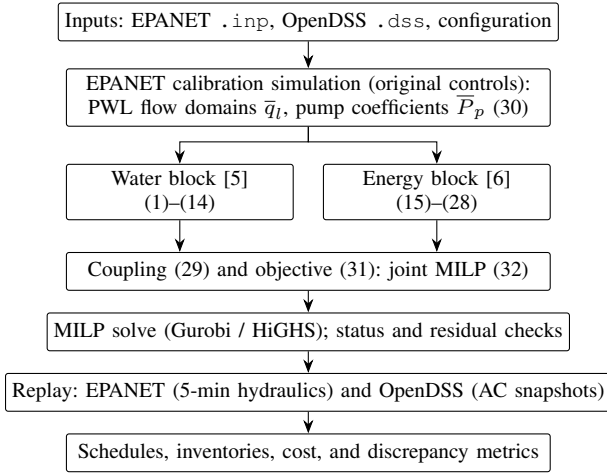


Fig. 2. Implementation workflow. Calibration and diagnostic linearizations are separated from the piecewise-linear scheduling formulation, and MILP feasibility is separated from nonlinear replay.

B. Comparison Protocol

The experiment asks whether choosing pump operation jointly with electrical dispatch changes the procurement cost relative to a feasible pumping schedule selected without regard to the electrical system, under the same asset availability and tariff. The independent baseline is constructed in two steps: the water block alone is solved with a uniform pump energy price, which yields an energy-efficient schedule as a water operator would select without price information; that schedule is then fixed in the joint model and the electrical dispatch is optimized under the time-varying prices. Joint scheduling releases the pump binaries and optimizes the same objective with identical network, asset, initial-inventory, and terminal constraints. A flat-price joint case serves as a tariff sensitivity. Only within-tariff comparisons with identical constraints estimate the scheduling benefit; the difference between bills under two tariffs combines an operating response with a price change, and a comparison with and without a battery measures both an asset change and its scheduling. A uniform-price water objective can admit several optimal pumping schedules, so the reported benefit is relative to the particular recorded baseline schedule, not an average over uncoordinated policies.

C. Replay

Solver feasibility, formulation tests, and physical replay answer different questions. Unit and integration tests establish that the constraints conserve inventory, enforce asset limits, preserve units, and route the pump demand to the correct bus. Algebraic residuals of the solved MILP establish consistency with the implemented surrogate. Replaying the optimized decisions in independent nonlinear simulators assesses physical fidelity on the instance. In the replay, the optimized pump statuses are imposed on the original EPANET network with a five-minute hydraulic time step and hourly reports, and the optimized net bus injections, including PV, storage, and pump power, are solved as balanced constant-power snapshots in OpenDSS. Agreement obtained after fixing variables at a

TABLE II
CASE-STUDY DATA (SYNTHETIC PROFILES, ONE-HOUR INTERVALS)

Item	Value
Water network	SNET: 1 reservoir, 1 pump, 2 junctions, 2 pipes, 1 tank
Tank	12 m diameter, elevation 25 m, level 0–6 m, initial 3 m
Demand, pressure floor	25 L/s nominal with daily pattern; 20 m
Pump	Design point 50 L/s at 40 m; $\eta_p = 0.75$; $\bar{P}_p = 25.948$ kW
PWL segments	12 (pipes, signed domain), 6 (pump), 5 (line currents)
Electrical network	Two-bus 115 kV circuit, 1000 kW nominal load
PV	100 kW at the load bus, availability peaking at hour 12
Battery	200 kWh, 50 kW charge/discharge, $\eta = 0.95$, $e^0 = 100$ kWh
Voltage band	$0.9 \leq v \leq 1.1$ p.u.
Purchase price	0.08 (h 0–6, 21–23), 0.16 (h 7–15), 0.32 (h 16–20)
Sale price	0

simulator's operating point would only be a consistency check; the replay reported here assesses the proposed schedules themselves.

VIII. CASE STUDY

A. Illustrative Case and Comparison Protocol

The water network is the supplied SNET example with one reservoir, one fixed-speed pump, two junctions, two pipes, and one cylindrical tank (Table II). The tank has a 12 m diameter, 25 m elevation, water-level bounds of 0–6 m, and an initial level of 3 m. Nominal demand is 25 L/s with the supplied 24-hour pattern; a 20 m junction-pressure floor and terminal tank replenishment are enforced. The pump design point is 50 L/s at 40 m. Pipe head losses use 12 segments over each signed flow domain and the pump curve six segments. The electrical network is the supplied two-bus 115 kV circuit with a 1000 kW nominal load and its daily multipliers; grid exchange is available at the source bus only. This lightly loaded circuit is an illustrative input, not a representative low-voltage feeder or a congestion stress test.

One 100 kW PV installation and one 200 kWh battery are located at the load bus. The battery has 50 kW charge and discharge limits, efficiencies of 0.95, and an initial and minimum terminal energy of 100 kWh. The synthetic purchase prices are 0.08 for hours 0–6 and 21–23, 0.16 for hours 7–15, and 0.32 for hours 16–20, in currency units per kWh; exports earn nothing. These profiles are controlled experimental inputs, not observations or an actual utility tariff. With a wire-to-water efficiency of 0.75 the calibrated on-state pump power from (30) is 25.948 kW.

The independent baseline first minimizes the water block alone under a uniform pump energy price, then fixes that pump schedule while optimizing the electrical dispatch under the time-varying prices. Joint scheduling releases the pump binaries and optimizes the same cost with identical network, asset, and terminal constraints. Because the uniform-price water problem can have several optimal schedules, the comparison

TABLE III
COMPUTED SCHEDULES. COSTS ARE IN SYNTHETIC CURRENCY UNITS;
THE FLAT-PRICE CASE USES A DIFFERENT TARIFF. SOLVE TIME IS
WALL-CLOCK SOLVER-CALL TIME ON THE RECORDED MACHINE

Case	Cost	On (h)	Import (kWh)	Peak (kW)	Solve (s)
Independent	2943.26	9	16954.05	920.0	0.07
Joint	2941.18	9	16954.05	915.9	4.25
Joint, flat price	1693.35	9	16933.53	960.0	2.66

measures the improvement over this explicit reproducible baseline rather than a universal bound on the value of coordination. A separate flat-price joint case (0.10 per kWh in every hour) is a tariff sensitivity and is not directly comparable in monetary terms.

B. Optimization Outcomes

Table III reports the saved solver outputs. Joint scheduling reduces the daily electricity procurement cost from 2943.26 to 2941.18 relative to the independent baseline, a difference of 2.08 (0.07%). Both schedules operate the pump for 9 hours and import the same energy, 16954 kWh; the saving comes entirely from timing (Fig. 3). Relative to the baseline, the joint schedule drops pumping in hours 0, 5, and 17 and adds hours 6, 8, and 13; at 25.948 kW per on-hour this tariff exchange is worth $25.948 \times (0.48 - 0.40) = 2.08$. The peak grid import changes from 920.0 to 915.9 kW. Both initial reserves are preserved in the optimization: the terminal tank heads are 28.04, 28.07, 28.19 m against an initial head of 28 m, and the terminal battery energy equals its initial 100 kWh. The available PV energy (700 kWh) is dispatched in full in every case. Under the time-varying tariff the battery charges 210.5 kWh and discharges 190 kWh; under the flat price it is not cycled at all, because without price differences the round-trip losses make storage arbitrage unprofitable. All values refer to the MILP approximation.

The joint model has 3219 variables and 2598 constraints. All three coupled solves and the independent water-only solve terminated as optimal at the 10^{-4} gap tolerance. The water-only solve took 0.43 s. The run directory records full-precision dispatch, inventory trajectories, solver logs, algebraic residuals (maximum constraint violation below 5×10^{-6}), the input configuration, dependency versions, and input and source hashes. Timing includes the Pyomo interface overhead and is not an isolated solver benchmark.

C. Independent Nonlinear Replay

Each optimized pump schedule is replayed in EPANET with a five-minute hydraulic time step and hourly reports, the original level controls being replaced by the hourly commands. The optimized net bus injections are replayed as balanced constant-power snapshots in OpenDSS, which retains the nonlinear AC behavior of the original circuit whereas the MILP neglects series losses and shunt effects. The electrical replay keeps the fixed on-state pump-power coefficient and therefore does not validate the pump energy calibration. Table IV summarizes the discrepancies.

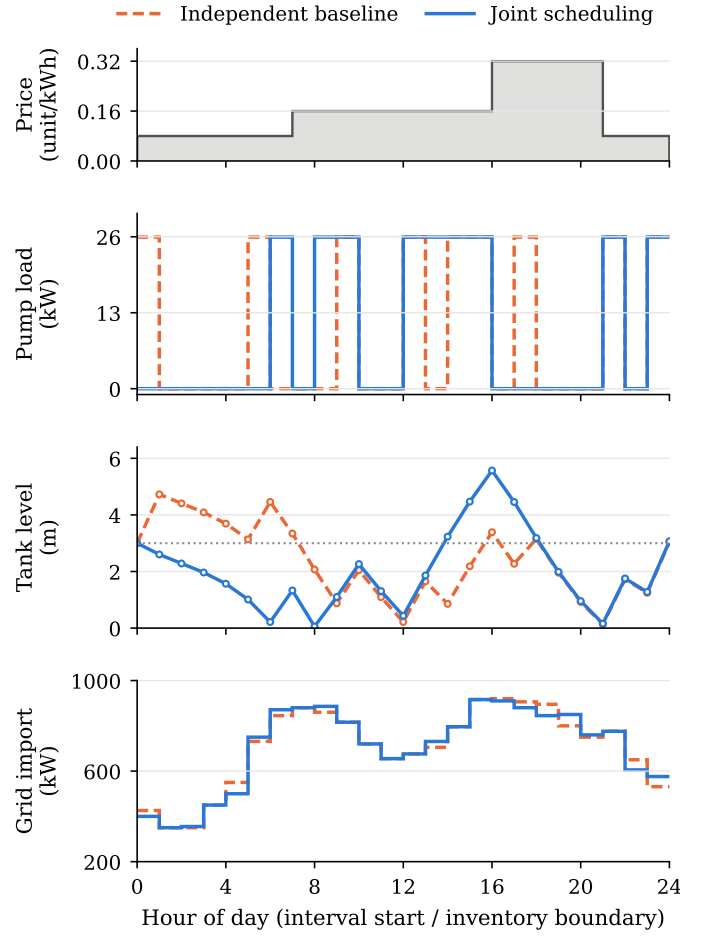


Fig. 3. Purchase price and optimized schedules for the independent baseline and joint scheduling: pump electrical load, tank water level, and grid import. Tank markers are inventory boundaries; power values apply over the following hourly interval. The dotted line marks the initial level, which the terminal level must reach.

TABLE IV
REPLAY OF THE OPTIMIZED SCHEDULES IN EPANET AND OPENDSS

Metric	Independent	Joint
Minimum junction pressure (m)	25.06	25.07
Pressure violations at report times	0	0
Pump-status mismatches at report times	0	0
Maximum tank-head discrepancy (m)	0.154	0.135
Maximum flow discrepancy (L/s)	0.466	0.437
Terminal tank-head change vs. initial (m)	-0.114	-0.067
All OpenDSS snapshots converged	yes	yes
Maximum voltage discrepancy (p.u.)	0.00034	0.00034
Voltage violations	0	0
Peak line loading (%)	1.31	1.29
Maximum grid-import discrepancy (kW)	0.005	0.005

Both schedules satisfy the pressure floor and the voltage band at every report time, and the pump statuses imposed in EPANET match the MILP decisions. Tank-head and flow discrepancies remain below 0.16 m and 0.47 L/s. The electrical discrepancies are small because the feeder is lightly loaded and the circuit is short, which is the regime in which the flat-voltage linearization is most accurate. The replay, however, ends with the tank 0.114 m (independent) and 0.067 m

(joint) below its initial level despite the MILP replenishment constraint (9). The hourly hydraulic approximation therefore does not preserve the terminal requirement exactly, and a reserve margin or a simulation-informed refinement of (7) is needed before operational use. Hourly report checks do not certify feasibility between reports. These results support the formulation as a scheduling prototype with explicit simulation checks; they are not a certification for field operation.

IX. DISCUSSION AND LIMITATIONS

The computed cost difference arises from timing rather than from reduced electricity use: the independent and joint schedules have the same total import and the same pump on-hours, and the joint schedule removes the single pumping hour that the baseline placed in the peak tariff block. The small relative saving should be read against the modest pump load compared with the feeder's non-pump demand and against the lightly loaded feeder, which does not let the experiment demonstrate congestion relief. The absolute mechanism, however, is exactly what the formulation is meant to capture: within the hydraulic feasible set defined by the water block, pumping moves to the hours in which the energy block finds electricity cheapest.

The fixed on-state pump coefficient (30) is the principal coupling approximation. It keeps the interface linear and makes the effect of pump timing directly visible, but it can favor schedules whose simulated flow and head depart from the calibration trajectory, and it does not reward operating a pump near its best-efficiency point. Modeled pump-energy savings are savings under that coefficient until replay flows and heads are converted to electrical consumption; the electrical replay in this study keeps the same coefficient and therefore assesses the feeder approximation, not the pump-energy model. A validated PWL relation between electrical pump power and the hydraulic operating point is the natural next step, and variable-speed pumps would require speed-dependent characteristics, both of which MILPNet's device structure can accommodate.

The terminal reserve shortfall observed in the hydraulic replay is a second consequence of the hourly discretization. Tank dynamics (7) constrain interval-end volumes with interval-average flows, whereas the simulator resolves within-hour head changes and the resulting variation of pump flow along its curve. A reserve margin above h_k^0 , a shorter interval, or a simulation-informed correction of the tank equation would close the gap; whichever is chosen, all compared schedules must share it.

Further limitations bound the numerical evidence. A small synthetic water network and a two-bus circuit do not establish scalability, performance under congestion, or behavior on real operator data. Coarse hydraulic interpolation can bias pressures and volumes, and flow domains derived from a reference schedule can exclude feasible operation outside that domain. The linearized power flow is most defensible near its flat-voltage expansion point, and the conservatism of (28) does not compensate for its error. Hourly decisions miss sub-hourly tank-control events, motor starts, and short voltage excursions. Water quality, leakage, battery wear, pump switching costs, equipment outages, and forecast uncertainty are absent.

Finally, the centralized decision structure assumes that one operator owns both systems; where the water and energy operators are distinct entities with private information, (32) is the benchmark that a decentralized coordination mechanism should approach, not a description of how such operators would actually interact.

X. CONCLUSION

This paper formulated the day-ahead coordinated scheduling of a water distribution network and a distributed energy system as a single MILP assembled from two published component models: the MILPNet water-network framework and the electrical part of the DES model of Morvaj *et al.*, with the heat balance omitted because the systems interact only through electrical demand. A single linear coupling constraint makes each pump's status an electrical load at its bus, so pumping is priced once through grid procurement and the problem class is unchanged. Consistent interval-end inventories and terminal replenishment for both tanks and batteries make the resulting costs interpretable across schedules. In the reproducible case study, joint scheduling lowered the procurement cost relative to an independently selected pumping schedule by moving pump operation out of the peak tariff block, with unchanged import energy. Nonlinear replay preserved sampled pressure and voltage limits but revealed a terminal water-reserve shortfall, which limits the operational interpretation of the MILP schedules until the hydraulic approximation is refined. Future work will replace the fixed pump coefficient by a validated flow- and head-dependent power model, test the formulation on larger networks and congested feeders, add forecast uncertainty, and use the centralized problem as the benchmark for decentralized coordination between separate water and energy operators.

DATA AND CODE AVAILABILITY

Network inputs, configuration, optimization code, tests, and the experiment scripts that regenerate Section VIII and Fig. 3 are included in the accompanying repository. From the repository root, `python -m src.experiments regenerate` the study from `data/inputs/paper_case.yaml`, and `latexmk -pdf main.tex` in the `paper` directory rebuilds the manuscript. The bundled MILPNet reference code is attributed to Thomas and Sela [5] and is distinct from the project implementation. No confidential measurements or external market data are required.

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